

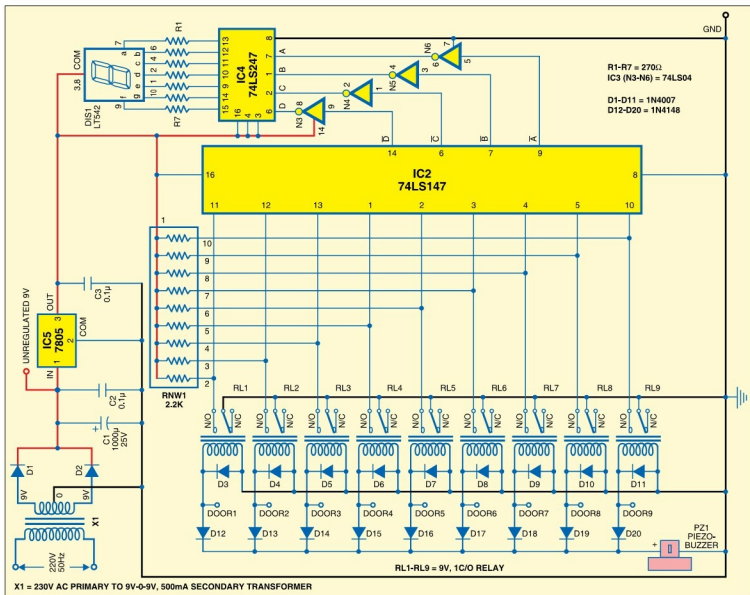


Fig. 2: Circuit of the control unit for multichannel door alarm with display

and connected to piezobuzzer PZ1. Diodes D3 through D11 are used as free-wheeling diodes.

To derive the power supply, 230V AC mains is stepped down by centre-tapped transformer X1 to deliver the secondary output of 9V-0-9V, 500mA. The transformer's output is rectified by a full-wave rectifier comprising diodes D1 and D2, filtered by capacitors C1 and C2, and regulated by IC 7805 (IC5). Capacitor C3 bypasses any ripple present in the regulated power supply. Unregulated 9V is used for the door units, relays and buzzer.

Initially, when the power supply is switched on and all the nine doors are closed, the inputs of priority encoder IC2 are held high via resistor

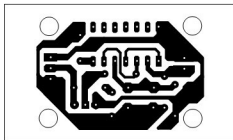


Fig. 3: Actual-size, single-side PCB for the door unit of multichannel door alarm with display

network RNW1, respectively, and its BCD output is '0000'. The BCD output of IC2 is applied to 7-segment decoder driver IC4 after inversion by hex inverter gates inside IC3. Thus when power supply is switched on, the display shows '0' without a buzzer sound.

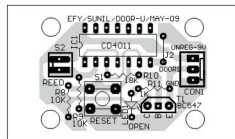


Fig. 4: Component layout for the PCB shown in Fig. 3

When any of the nine doors, say, door 1, is opened, the corresponding transistor (T1) is driven into saturation and relay RL1 energises. The corresponding input pin of 74LS147 (which is connected to the contacts of relay RL1) goes low and the display shows the number of